

$$\begin{pmatrix} 3 & 3 & 5 & 9 & 17 \\ 3 & 5 & 9 & 17 & 33 \\ 5 & 9 & 17 & 33 & 65 \\ 9 & 17 & 33 & 65 & 129 \\ 17 & 33 & 65 & 129 & 257 \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \\ a_4 \end{pmatrix} = \begin{pmatrix} 0 \\ -4 \\ -10 \\ -22 \\ -46 \end{pmatrix}$$

System mi vypočítal řešení:

$$a_0 = 1 \quad a_1 = \frac{4}{5} \quad a_2 = \frac{3}{5} \quad a_3 = \frac{1}{5} \quad a_4 = -\frac{3}{5}$$

Pro ověření jsem dosadil a opravdu dostáváme pravou stranu, kterou chceme.

$$\frac{15}{5} + \frac{12}{5} + \frac{15}{5} + \frac{9}{5} - \frac{51}{5} = \frac{1}{5}(51-51) = 0$$

$$\frac{15}{5} + \frac{20}{5} + \frac{17}{5} + \frac{17}{5} - \frac{99}{5} = \frac{1}{5}(35+44-99) = \frac{-20}{5} = -4$$

$$\frac{25}{5} + \frac{36}{5} + \frac{51}{5} + \frac{33}{5} - \frac{195}{5} = \frac{1}{5}(112+33-195) = \frac{-50}{5} = -10$$

$$\frac{45}{5} + \frac{68}{5} + \frac{99}{5} + \frac{65}{5} - \frac{387}{5} = \frac{1}{5}(113+164-387) = \frac{1}{5}(277-387) = \frac{-110}{5} = -22$$

$$\frac{85}{5} + \frac{132}{5} + \frac{195}{5} + \frac{129}{5} - \frac{771}{5} = \frac{1}{5}(280+261-771) = \frac{541-771}{5} = \frac{-230}{5} = -46$$

Polynom tedy je: $-\frac{3}{5}x^4 + \frac{1}{5}x^3 + \frac{3}{5}x^2 + \frac{4}{5}x + 1$